

HACKATHON PROBLEM STATEMENTS

Elevate 1.0

Dwarkadas J. Sanghvi College of Engineering, Vile Parle (W) · Institution's Innovation Council, DJSCE

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PROBLEM STATEMENT 01

Seamless Aviation Journey Experience

Background

Air travel is no longer a single transaction of booking a seat and boarding a plane. It is a continuous experience that begins when a person decides to travel and extends through planning, preparation, departure, the airport, the flight, connections, arrival, and the journey back home. Travellers must discover destinations, construct itineraries around dates, budgets, companions and personal constraints, compare the true cost and trade-offs of different choices, understand airlines and airports, manage documents and travel requirements, navigate terminals, track flights and baggage, handle connections and disruptions, coordinate with other travellers, use offers and loyalty benefits, arrange accommodation and ground transportation, and transition into the destination.

Yet the ecosystem supporting this experience remains deeply fragmented. Airlines, airports, booking platforms, hotels, transport providers, card and loyalty programmes, financial offers, government and regulatory systems, and local services operate across separate experiences. Information, rules, eligibility conditions, prices, benefits, documents and updates are scattered across different sources. The traveller is left to gather this information, determine what is relevant, compare alternatives, understand consequences, coordinate multiple bookings and people, and continuously adjust their plans—often while dealing with uncertainty and time pressure. The same journey can also be radically different for a first-time flyer, frequent traveller, family, business traveller, elderly passenger, or traveller with accessibility requirements.

Problem Statement

The challenge is to transform the aviation journey into a seamless, intelligent, mobile-first experience that connects the traveller's needs across the entire journey, from planning and booking to airport navigation, flight management, disruptions, and arrival. The solution should help travellers make informed decisions by considering factors such as total journey cost, time, convenience, reliability, baggage, and personal constraints. It should also contextualize card offers, loyalty benefits, rewards, and discounts against the traveller's actual journey, rather than presenting them as isolated promotions.

The experience must extend beyond booking to provide relevant, actionable guidance throughout the journey. Travellers should be able to navigate airports, understand requirements, track flights and baggage, manage connections, and access the right information at the right time without switching between disconnected services.

Since travel conditions can change unexpectedly, the system should account for delays, cancellations, missed connections, gate changes, and other disruptions. It should identify their potential impact on the wider itinerary, including accommodation and ground transportation, and help travellers adapt their plans accordingly. A key technical challenge is balancing on-device and cloud AI. Essential capabilities such as accessing travel documents, boarding passes, offline guidance, and translation should remain useful with limited connectivity, while computation-intensive or live-data-dependent tasks, such as itinerary planning, disruption management, and offer comparison, can leverage the cloud. The experience should degrade gracefully offline while considering latency, battery usage, privacy, and computational cost.

The solution must also adapt to different traveller profiles, including first-time flyers, frequent travellers, families, groups, business travellers, and people with accessibility or other specific requirements. Multilingual interaction must be built into the experience, supporting regional languages such as Hindi, Marathi, Tamil, and Bengali, as well as mixed-language input such as Hinglish, through text, voice, and on-screen content.

Throughout, UI/UX is a core priority. The experience must present complex information clearly on a mobile screen, minimize unnecessary steps, surface the most relevant next action, and remain intuitive even when travellers are unfamiliar with their surroundings or under time pressure. The objective is not simply to combine existing services, but to make the entire aviation experience more coherent, personalized, and effortless.

CORE QUESTION

How can the complete aviation experience – from discovering and planning a journey to navigating airports, managing flights and disruptions, accessing services and benefits, reaching the destination, and returning home – be transformed into a seamless, intuitive and effortless mobile experience for every traveller, in their own language?

PROBLEM STATEMENT 02

Racket-Sports Intelligence from One Camera

Background

Professional racket sports – tennis, badminton, table tennis, squash, padel and pickleball – have access to sophisticated tracking, officiating, player analytics, and performance systems, but this infrastructure is largely unavailable outside elite competition. At club, academy, district, and state levels, matches are often conducted with little or no structured digital record. Scores may be inconsistently maintained, ball or shuttle speed and trajectory are unavailable, player statistics are scattered, disputes can be difficult to resolve, and historical match information is rarely preserved in a usable form. Player profiles, performance history, rankings, and development data are similarly fragmented across coaches, academies, tournaments, and individuals. As a result, a large part of the racket-sports ecosystem operates without the data infrastructure required for objective analysis, transparent competition, and continuous player development.

Problem Statement

Professional racket sports have access to sophisticated tracking, officiating, player analytics, and performance systems, but this infrastructure is largely unavailable outside elite competition. At club, academy, district, and state levels, matches are often conducted with little or no structured digital record. Scores may be inconsistently maintained, ball or shuttle speed and trajectory are unavailable, player statistics are scattered, disputes can be difficult to resolve, and historical match information is rarely preserved. Player profiles, performance history, rankings, and development data are similarly fragmented across coaches, academies, tournaments, and individuals.

This fragmentation extends beyond the match itself. Players lack accessible ways to understand their performance, compare themselves with others, find suitable opponents, organise meetups, discover tournaments, or find and book courts. Coaches lack continuous data to evaluate players and guide their development, while local racket-sports communities remain disconnected. There is little accessible intelligence for analysing matches, learning from historical performance, generating predictions, or providing personalised coaching and feedback through mobile, offline, or voice-based experiences.

Solutions should work across racket sports rather than a single game, handling the differences in court dimensions, scoring rules, and the size and speed of the ball or shuttle, using only one normal camera such as a phone. No training should be required: the system must work out of the box on a new court, venue or sport without collecting labelled data or training or fine-tuning models for it. Packaging these capabilities as an SDK and API so that developers, academies and tournament organisers can build on them in the future is a brownie point, as is an approach that also generalises beyond racket sports to other sports.

The challenge is therefore to rethink how racket sports can become more connected, measurable, intelligent, and accessible – from the individual player and coach to matches, tournaments, courts, and the wider racket-sports community.

CORE QUESTION

How much of a professional tracking and intelligence system can be rebuilt from one normal camera, across racket sports and without any training, while managing and connecting players?

PROBLEM STATEMENT 03

An Evolving AI Research Environment

Background

Machine learning research increasingly spans papers, datasets, code, notebooks, compute platforms, documentation, experiments, collaborators, and countless disconnected applications. Reproducing published work is difficult because implementation details, dependencies, configurations, preprocessing, datasets, and reported results may differ across sources and environments. Even after an experiment runs, researchers repeatedly face debugging, dependency conflicts, data issues, hyperparameter choices, model comparisons, failed experiments, and unanswered “what if?” questions. Valuable knowledge from these successes and failures is rarely retained in a way that helps future research. As research scales, the researcher spends increasing time managing the machinery around experimentation rather than investigating the research itself.

AI agents are increasingly capable of reading papers, writing and debugging code, and running experiments. However, most agentic systems still depend on predefined agents, fixed roles, manually designed workflows, and known task boundaries, while every paper and research question needs a different mix of skills. A research environment built on agents therefore requires more than capable individual agents – it requires a system that can dynamically organise, coordinate, evaluate, and evolve its agents around the research itself.

Problem Statement

Build an intelligent research environment, powered by a dynamic team of AI agents, that can support the researcher across the complete lifecycle of machine learning research. The system should be capable of taking a paper, research question, algorithm, dataset, codebase, or experimental idea and working with the information required to turn it into meaningful experimentation. It should be able to execute research across environments such as local machines, Kaggle, Colab, remote servers, or other available compute, while handling the practical problems that arise during execution including dependencies, incompatible environments, data issues, implementation errors, resource constraints, configuration choices, and hyperparameter decisions.

Rather than relying on a fixed set of predefined agents, a meta-controller should determine the capabilities each research task requires – such as reading papers, implementation, environment setup, debugging, experiment design, and analysis – coordinate existing agents, and spawn or adapt new specialised agents when existing capabilities are insufficient. Agents should be evaluated through a meaningful reward loop based on research outcomes, compute cost, and researcher feedback. Agents that consistently produce valuable results should be refined, while those that fail to justify their cost or contribution should be merged, replaced, or retired.

The system should go beyond checking whether a paper can be reproduced. It should be capable of running experiments, observing results, diagnosing failures, repairing and improving implementations, comparing multiple papers or approaches under meaningful conditions, and determining why results differ from reported findings. When a researcher asks questions such as “what if this dataset is changed?”, “what if this architecture is modified?”, or “what happens under these conditions?”, the system should be capable of investigating multiple possibilities rather than requiring every experiment to be manually designed and executed.

Experiments should not exist as isolated runs. What the system learns from successful and failed executions, debugging decisions, configurations, comparisons, and outcomes should remain useful for future research. Over time, the system should become better at deciding what to try, how to execute it, how to recover when it fails, which direction is worth pursuing next, and which agents are best suited to each task.

The research environment should also bring together the surrounding work that normally lives across thousands of tools: discovering and reading papers, managing research notes and Markdown files, working with datasets and code, creating documents and research outputs, maintaining experimental records, and collaborating with other researchers and authors. A researcher should be able to control ongoing work remotely, including through a mobile interface, while longer-running computation and experimentation may continue elsewhere.

The challenge also extends to research autonomy. Multiple possible hypotheses, methods, datasets, or experimental conditions may need to be investigated simultaneously, with the system determining which directions deserve further exploration based on evidence and previous outcomes. The researcher should remain able to inspect, intervene, guide, approve, reject, or redirect this process.

The system should be demonstrated on modest compute, such as a laptop CPU or free-tier Kaggle and Colab, choosing model sizes, dataset subsets, and experiment budgets that still produce meaningful evidence. Participants should take at least one paper – for example from the research directions below, spanning materials science, edge and efficient ML, CPU-scale machine learning, and 3D – reproduce its key result autonomously, investigate at least one meaningful "what if?" extension, and explain where and why their results differ from the paper.

Finally, the system may contain a very large number of concurrent agents, activities, decisions, experiments, resources, documents, and research threads. How these are organised and exposed to the researcher is itself a major part of the challenge. The interface should make clear what the agents are doing, why they are doing it, how they are performing, and how the team is evolving, making a complex autonomous research environment understandable and controllable without reducing the underlying capability to a simple notebook or chat interface.

CORE QUESTION

How can you build a single environment where a researcher can conduct, control, and collaborate on their entire research process while an evolving team of AI agents continuously executes, investigates, learns, and improves alongside them?

Suggested Research Directions

Participants may choose any paper; the papers below are starting points that are feasible on a CPU or free-tier compute as supported by our sponsors.

Materials science

- CGCNN — Xie & Grossman, "Crystal Graph Convolutional Neural Networks for an Accurate and Interpretable Prediction of Material Properties" (Physical Review Letters, 2018)
- MEGNet — Chen et al., "Graph Networks as a Universal Machine Learning Framework for Molecules and Crystals" (Chemistry of Materials, 2019)
- Roost — Goodall & Lee, "Predicting materials properties without crystal structure: deep representation learning from stoichiometry" (Nature Communications, 2020)
- Matbench — Dunn et al., "Benchmarking materials property prediction methods: the Matbench test set and Automatminer reference algorithm" (npj Computational Materials, 2020)
- CHGNet — Deng et al., "CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling" (Nature Machine Intelligence, 2023), using the pretrained model on CPU

Edge and efficient ML

- Jacob et al., “Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference” (CVPR, 2018)
- Hinton, Vinyals & Dean, “Distilling the Knowledge in a Neural Network” (2015)
- Frankle & Carbin, “The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks” (ICLR, 2019)
- Sandler et al., “MobileNetV2: Inverted Residuals and Linear Bottlenecks” (CVPR, 2018)
- Lin et al., “MCUNet: Tiny Deep Learning on IoT Devices” (NeurIPS, 2020)

Machine learning on a CPU

- Chen et al., “SLIDE: In Defense of Smart Algorithms over Hardware Acceleration for Large-Scale Deep Learning Systems” (MLSys, 2020)
- Power et al., “Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets” (2022)
- Liu et al., “KAN: Kolmogorov–Arnold Networks” (2024)
- Raissi, Perdikaris & Karniadakis, “Physics-informed neural networks” (Journal of Computational Physics, 2019)
- Hollmann et al., “TabPFN: A Transformer That Solves Small Tabular Classification Problems in a Second” (ICLR, 2023)
- Grinsztajn, Oyallon & Varoquaux, “Why do tree-based models still outperform deep learning on tabular data?” (NeurIPS, 2022)

3D

- Qi et al., “PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation” (CVPR, 2017)
- Wang et al., “Dynamic Graph CNN for Learning on Point Clouds” (ACM Transactions on Graphics, 2019)
- Park et al., “DeepSDF: Learning Continuous Signed Distance Functions for Shape Representation” (CVPR, 2019), on a few shapes
- Mildenhall et al., “NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis” (ECCV, 2020), as a low-resolution “tiny NeRF”
- Fridovich-Keil et al., “Plenoxels: Radiance Fields without Neural Networks” (CVPR, 2022)

PROBLEM STATEMENT 04

AI-Run Statutory Revenue Audit

Background

A mid-size manufacturer records 6,000 to 8,000 sales a year, yet its statutory auditor can test only a few dozen. That testing still runs on spreadsheets. The client exports a sales register and a related-party list, the audit team copies them into a workpaper, picks samples on judgment, and chases invoices and transport documents over email. The auditor sees only what the client exports, never the systems that recorded what was promised, what physically moved and what was reversed after year end. Fake-invoice networks, circular trading, period-end stuffing and hidden related-party sales keep surfacing in enforcement actions, almost always behind paperwork that looked complete.

Problem Statement

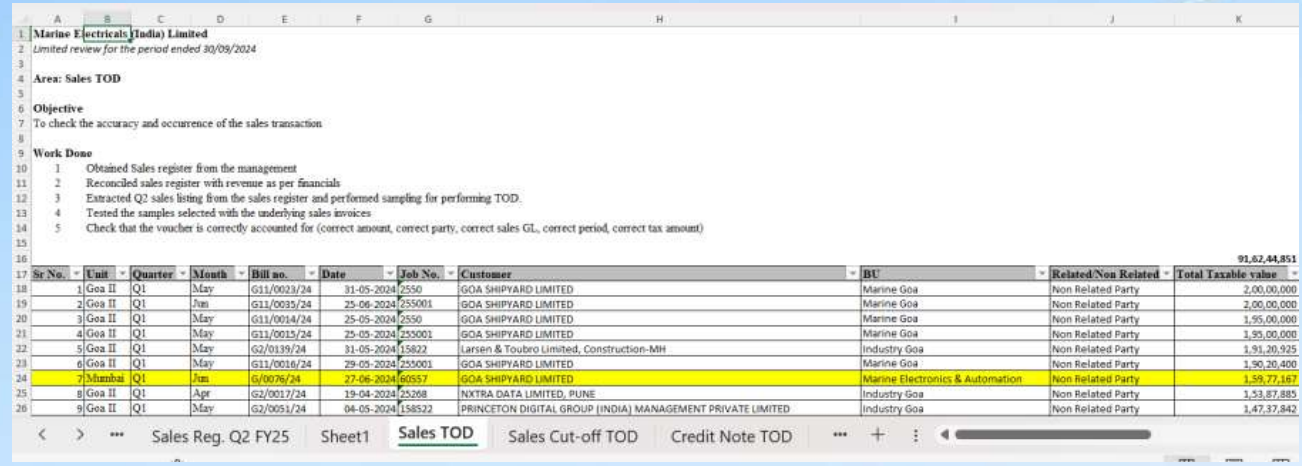
Core

The client uploads its sales ledger through a client interface, along with details of its related parties, and the system is notified of the upload. The ledger is carried into a new file that serves as the audit workpaper: the client's data is pasted exactly as provided, and each customer is marked as a related or non-related party by tracing it against the related-party details. Since 6,000 to 8,000 sales cannot all be tested, a separate Sales TOD sheet selects around 25 to 35 samples. The selection may be random but must account for risk, especially related parties, where the chance of bogus sales is high. It should cover as many areas as possible: different customers, high amounts, and related and non-related parties. A Sales Cut-off TOD takes roughly the last 10 invoices before 31 March and the first 10 to 15 after it, verifying the corresponding dispatch or delivery dates and accounting entries, since period-end sales are the most likely to be manipulated. The selected samples are sent to the client automatically, and the client uploads the supporting details for each one, such as the invoice, e-way bill and LR copy, from which testing begins.

Reference: what the auditor works with today

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
2	Bill no.	Date	Job No.	Customer	Ledger Acc	State Code	GST No.	Total Cost	Discount	Packing	Freight	Insurance	IGST	CGST	SGST	TCSCollection	Round	Gross Amt	Tax	BU	E Invoice	
3	S/0069/24	02-07-2024	607271	THE COMI	SALES AC	27		35,000	0	0	0	0	0	875	875	0	0	36,750	CGST/SGS	Marine Elec	Incomplete	
4	G15/00767	02-07-2024	42762	COCHIN S	SALES AC	36	36AAACCE	1,98,000	0	0	0	0	35,640	0	0	0	0	2,33,640	IGST 18%	Marine Mur	Complete	
5	G15/00777	05-07-2024	23087	HYDRODY	SALES AC	27	27AAACH2	12,100	0	0	0	0	0	1,089	1,089	0	0	14,278	CGST/SGS	Marine Mur	Complete	
6	S/0070/24	05-07-2024	4284	NAUTILUS	SALES AC	33	33AACCN5	42,000	0	0	0	0	7,560	0	0	0	0	49,560	IGST 18%	Marine Mur	Complete	
7	G/0087/24	08-07-2024	60655	GARDEN F	SALES AC	19	19AAACG6	4,36,566	0	0	0	0	21,828	0	0	0	0	4,58,394	IGST 5%	Marine Elec	Complete	
8	S/0071/24	09-07-2024	81211	INDIAN MA	SALES AC	33	33AAAA26	3,24,000	0	0	0	0	16,200	0	0	0	0	3,40,200	IGST 5%	Marine Elec	Complete	
9	G/0088/24	09-07-2024	81270	SEAMEC L	SALES AC	27	27AABCPE	50,000	0	0	0	0	0	4,500	4,500	0	0	59,000	CGST/SGS	Marine Mur	Complete	
10	S/0072/24	09-07-2024	40317	SEAMEC L	SALES AC	27	27AABCPE	25,503	0	0	0	0	0	2,295	2,295	0	0	30,094	CGST/SGS	Marine Mur	Complete	
11	G15/00787	09-07-2024	242612	DESMI INC	SALES AC	36	36AACD2	66,000	0	1,320	8,000	0	13,558	0	0	0	0	88,878	IGST 18%	Marine Mur	Complete	
12	G/0089/24	10-07-2024	81213	MARINE C	SALES AC	37	37AAIPA2	3,94,896	0	0	0	0	71,081	0	0	0	0	4,65,977	IGST 18%	Marine Elec	Complete	
13	G15/00794	10-07-2024	26665	HOOGHLY	SALES AC	19	19AAECH3	22,70,000	0	0	0	0	4,08,600	0	0	0	0	26,78,600	IGST 18%	Marine Mur	Complete	
14	G15/00800	10-07-2024	26651	HOOGHLY	SALES AC	19	19AAECH3	22,70,000	0	0	0	0	4,08,600	0	0	0	0	26,78,600	IGST 18%	Marine Mur	Complete	
15	G15/00817	11-07-2024	9188	CONTROL	SALES AC	27		14,13,162	0	0	0	0	0	35,329	35,329	0	0	14,83,820	CGST/SGS	Marine Mur	Incomplete	
16	G15/00822	11-07-2024	9562	DEEP INDI	SALES AC	24	24AACCD5	3,84,750	0	0	0	0	69,255	0	0	0	0	4,54,005	IGST 18%	Marine Mur	Complete	
17	G/0090/24	12-07-2024	607278	THE COMI	SALES AC	37		36,600	0	0	0	0	1,830	0	0	0	0	38,430	IGST 5%	Marine Elec	Incomplete	
18	S/0073/24	12-07-2024	607279	THE COMI	SALES AC	33		92,800	0	0	0	0	4,640	0	0	0	0	97,440	IGST 5%	Marine Elec	Incomplete	
19	S/0074/24	12-07-2024	607281	THE COMI	SALES AC	29		70,000	0	0	0	0	3,500	0	0	0	0	73,500	IGST 5%	Marine Elec	Incomplete	
20	G15/00833	12-07-2024	2667	ALEKTON	SALES AC	33	33AAACA3	5,11,350	0	15,116	11,500	0	96,834	0	0	0	0	6,34,799	IGST 18%	Marine Mur	Complete	
21	G/0091/24	12-07-2024	607264	THE SHIPS	SALES AC	27	27AAACT1	5,64,520	0	0	0	0	0	14,113	14,113	0	0	5,92,746	CGST/SGS	Marine Elec	Complete	
22	S/0075/24	12-07-2024	607276	PATEL EN	SALES AC	37	37AAFFP9	3,70,830	0	0	0	0	18,542	0	0	0	1	3,89,372	IGST 5%	Marine Elec	Complete	
23	CN/0008/2	15-07-2024	60515	MAZAGON	SALES AC	0		-2,50,000	0	0	0	0	0	-22,500	-22,500	0	0	-2,95,000	CGST/SGS	Marine Elec	Complete	
24	G/0092/24	15-07-2024	2661	COCHIN S	SALES AC	32	32AAACC6	14,68,000	0	0	0	0	2,64,240	0	0	0	0	17,32,240	IGST 18%	Marine Goo	Complete	
25	G15/00842	15-07-2024	22081	DEFENCE	SALES AC	36		24,88,447	0	28,500	45,000	0	1,28,097	0	0	0	0	26,90,044	IGST 5%	Marine Mur	Incomplete	
26	G15/00855	15-07-2024	2327	ENGIE AXI	SALES AC	27	27AAKCA5	5,21,710	0	0	0	0	0	46,954	46,954	0	0	6,15,618	CGST/SGS	Marine Mur	Complete	
27	G/0093/24	16-07-2024	60676	THE PROJ	SALES AC	37		6,38,191	0	0	0	0	31,910	0	0	0	0	6,70,101	IGST 5%	Marine Elec	Incomplete	
28	G15/00865	16-07-2024	2062	ALFA LAV	SALES AC	27	27AAACA5	2,75,000	0	0	0	0	0	24,750	24,750	0	0	3,24,500	CGST/SGS	Marine Mur	Complete	
29	G/0094/24	17-07-2024	81280	COCHIN S	SALES AC	32	32AAACC6	12,36,797	0	0	0	0	2,22,823	0	0	0	0	14,59,420	IGST 18%	Marine Elec	Complete	
30	S/0076/24	17-07-2024	607284	THE COMI	SALES AC	24		47,500	0	0	0	0	2,375	0	0	0	0	49,875	IGST 5%	Marine Elec	Incomplete	
31	S/0077/24	17-07-2024	607285	THE COMI	SALES AC	24		47,500	0	0	0	0	2,375	0	0	0	0	49,875	IGST 5%	Marine Elec	Incomplete	
32	G/0095/24	17-07-2024	81281	COCHIN S	SALES AC	32	32AAACC6	3,45,218	0	0	0	0	62,139	0	0	0	0	4,07,357	IGST 18%	Marine Elec	Complete	

Fig. 1: Extract of the client's sales register as received.



Marine Electricals (India) Limited

Limited review for the period ended 30/09/2024

Area: Sales TOD

Objective

To check the accuracy and occurrence of the sales transaction

Work Done

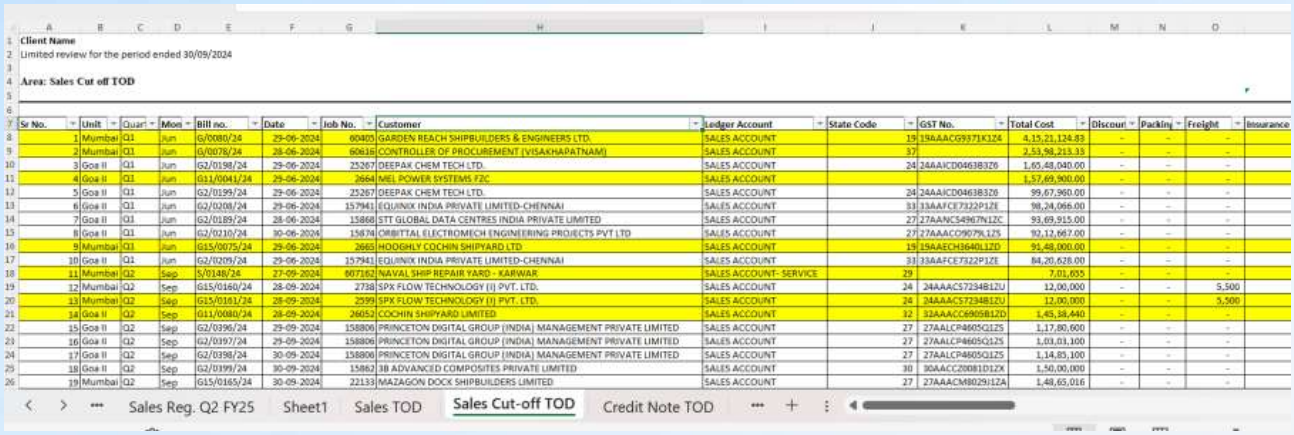
1. Obtained Sales register from the management
2. Reconciled sales register with revenue as per financials
3. Extracted Q2 sales listing from the sales register and performed sampling for performing TOD.
4. Tested the samples selected with the underlying sales invoices
5. Check that the voucher is correctly accounted for (correct amount, correct party, correct sales GL, correct period, correct tax amount)

91,62,44,851

Sr No.	Unit	Quarter	Month	Bill no.	Date	Job No.	Customer	BU	Related/Non Related	Total Taxable value
1	Goa II	Q1	May	G11/0023/24	31-05-2024	2550	GOA SHIPYARD LIMITED	Marine Goa	Non Related Party	2,00,00,000
2	Goa II	Q1	Jun	G11/0035/24	25-06-2024	255001	GOA SHIPYARD LIMITED	Marine Goa	Non Related Party	2,00,00,000
3	Goa II	Q1	May	G11/0014/24	25-05-2024	2550	GOA SHIPYARD LIMITED	Marine Goa	Non Related Party	1,95,00,000
4	Goa II	Q1	May	G11/0015/24	25-05-2024	255001	GOA SHIPYARD LIMITED	Marine Goa	Non Related Party	1,95,00,000
5	Goa II	Q1	May	G2/0139/24	31-05-2024	15822	Larsen & Toubro Limited, Construction-MH	Industry Goa	Non Related Party	1,91,20,925
6	Goa II	Q1	May	G11/0016/24	29-05-2024	255001	GOA SHIPYARD LIMITED	Marine Goa	Non Related Party	1,90,20,400
7	Mumbai	Q1	Jun	G/0076/24	27-06-2024	60557	GOA SHIPYARD LIMITED	Marine Electronics & Automation	Non Related Party	1,59,77,167
8	Goa II	Q1	Apr	G2/0017/24	19-04-2024	25268	NXTRA DATA LIMITED, PUNE	Industry Goa	Non Related Party	1,53,87,885
9	Goa II	Q1	May	G2/0051/24	04-05-2024	158522	PRINCETON DIGITAL GROUP (INDIA) MANAGEMENT PRIVATE LIMITED	Industry Goa	Non Related Party	1,47,37,842

Sales Reg. Q2 FY25 | Sheet1 | Sales TOD | Sales Cut-off TOD | Credit Note TOD

Fig. 2: Sales TOD sheet with objective, work done and selected samples.



Client Name

Limited review for the period ended 30/09/2024

Area: Sales Cut off TOD

Sr No.	Unit	Quarter	Month	Bill no.	Date	Job No.	Customer	Ledger Account	State Code	GST No.	Total Cost	Discount	Packing	Freight	Insurance
1	Mumbai	Q1	Jun	G/0080/24	29-06-2024	60405	GARDEN REACH SHIPBUILDERS & ENGINEERS LTD.	SALES ACCOUNT	19	19AAACG9371K1Z4	4,15,21,124.83	-	-	-	
2	Mumbai	Q1	Jun	G/0078/24	28-06-2024	60616	CONTROLLER OF PROCUREMENT (VISAKHAPATNAM)	SALES ACCOUNT	37		2,53,98,213.33	-	-	-	
3	Goa II	Q1	Jun	G2/0198/24	29-06-2024	25267	DEEPAK CHEM TECH LTD.	SALES ACCOUNT	24	24AAICD0463B3Z6	1,65,48,040.00	-	-	-	
4	Goa II	Q1	Jun	G11/0041/24	29-06-2024	2664	MEL POWER SYSTEMS FZC	SALES ACCOUNT			1,57,69,900.00	-	-	-	
5	Goa II	Q1	Jun	G2/0199/24	29-06-2024	25267	DEEPAK CHEM TECH LTD.	SALES ACCOUNT	24	24AAICD0463B3Z6	99,67,960.00	-	-	-	
6	Goa II	Q1	Jun	G2/0208/24	29-06-2024	157941	EQUINIX INDIA PRIVATE LIMITED-CHENNAI	SALES ACCOUNT	33	33AAFCE7322P1ZE	98,24,066.00	-	-	-	
7	Goa II	Q1	Jun	G2/0189/24	28-06-2024	15868	STT GLOBAL DATA CENTRES INDIA PRIVATE LIMITED	SALES ACCOUNT	27	27AANCS4967N1ZC	93,69,915.00	-	-	-	
8	Goa II	Q1	Jun	G2/0210/24	30-06-2024	15874	ORBITTAL ELECTROMECH ENGINEERING PROJECTS PVT LTD	SALES ACCOUNT	27	27AAACO9079L1Z5	92,12,667.00	-	-	-	
9	Mumbai	Q1	Jun	G15/0075/24	29-06-2024	2665	HOOGHLY COCHIN SHIPYARD LTD	SALES ACCOUNT	19	19AAECH3640L1ZD	91,48,000.00	-	-	-	
10	Goa II	Q1	Jun	G2/0209/24	29-06-2024	157941	EQUINIX INDIA PRIVATE LIMITED-CHENNAI	SALES ACCOUNT	33	33AAFCE7322P1ZE	84,20,628.00	-	-	-	
11	Mumbai	Q2	Sep	S/0148/26	27-09-2024	607162	NAVAL SHIP REPAIR YARD - KARWAR	SALES ACCOUNT- SERVICE	29		7,01,655	-	-	-	
12	Mumbai	Q2	Sep	G15/0160/24	28-09-2024	2738	SPX FLOW TECHNOLOGY (I) PVT. LTD.	SALES ACCOUNT	24	24AAACS7234B1ZU	12,00,000	-	-	5,500	
13	Mumbai	Q2	Sep	G15/0161/24	28-09-2024	2599	SPX FLOW TECHNOLOGY (I) PVT. LTD.	SALES ACCOUNT	24	24AAACS7234B1ZU	12,00,000	-	-	5,500	
14	Goa II	Q2	Sep	G11/0080/24	28-09-2024	26052	COCHIN SHIPYARD LIMITED	SALES ACCOUNT	32	32AAACC6905B1ZD	1,45,38,440	-	-	-	
15	Goa II	Q2	Sep	G2/0396/24	29-09-2024	158806	PRINCETON DIGITAL GROUP (INDIA) MANAGEMENT PRIVATE LIMITED	SALES ACCOUNT	27	27AALCP4605Q1Z5	1,17,80,600	-	-	-	
16	Goa II	Q2	Sep	G2/0397/24	29-09-2024	158806	PRINCETON DIGITAL GROUP (INDIA) MANAGEMENT PRIVATE LIMITED	SALES ACCOUNT	27	27AALCP4605Q1Z5	1,03,03,100	-	-	-	
17	Goa II	Q2	Sep	G2/0398/24	30-09-2024	158806	PRINCETON DIGITAL GROUP (INDIA) MANAGEMENT PRIVATE LIMITED	SALES ACCOUNT	27	27AALCP4605Q1Z5	1,14,83,100	-	-	-	
18	Goa II	Q2	Sep	G2/0399/24	30-09-2024	15862	3B ADVANCED COMPOSITES PRIVATE LIMITED	SALES ACCOUNT	30	30AACC2008D1Z9	1,50,00,000	-	-	-	
19	Mumbai	Q2	Sep	G15/0165/24	30-09-2024	22133	MAZAGON DOCK SHIPBUILDERS LIMITED	SALES ACCOUNT	27	27AAACM8029J1ZA	1,48,65,016	-	-	-	

Sales Reg. Q2 FY25 | Sheet1 | Sales TOD | Sales Cut-off TOD | Credit Note TOD

Fig. 3: Sales Cut-off TOD sheet with invoices around the period end.

Beyond the Core

The register is only a report of the business, so solutions must also build the system of record behind it and audit that instead. This covers the full commercial life of a sale: first customer interaction, commitment and pricing, physical movement of goods, revenue and tax recognition, cash settlement, and the promotions that drive demand. Everything must post to a double-entry ledger that cannot be silently altered. No schema or module breakdown is given; deciding what must exist to prove a sale really happened is part of the problem. The whole system must be exposed as a Model Context Protocol server, so that an AI agent holding only an identity can run the business as a client employee or audit it as an independent auditor, with exactly that identity's permissions. An auditor identity must never change business data, and any client change made after testing begins must be detected and traced to the affected samples.

The engine will be judged on what it catches. Every difference between the exported register and the system of record is a finding, not a cleaning step. Related parties may be undisclosed, may begin partway through the period, or may be linked only through shared identifiers. Period allocation follows transfer of control, not invoice date. Turnover with no physical movement behind it, period-end concentration of sales and discounts, inconsistencies across statutory documents and tax filings, revenue entries that did not originate from a sale, and evidence that is forged or belongs to another transaction must all surface. Credit notes, especially those issued after the period end, are tested too. Every sample selection must be reproducible and carry its reasoning. The planted irregularities will not be disclosed, and judges will exercise the system live through their own agent with no code changes allowed. The output is a workpaper an engagement

partner would sign (Sales Register, Sales TOD, Sales Cut-off TOD, Credit Note TOD), a findings memo with impact and proposed adjustments, and a tamper-evident audit trail.

Solutions should be grounded in the Standards on Auditing and the applicable accounting and indirect-tax framework, with a human auditor confirming every conclusion. Continuous auditing, multi-entity consolidation, a counterparty graph exposing cycles, regional-language document handling, plain-language justification of any finding, and an on-premise mode are brownie points, as is extending the approach to purchases, inventory or payroll.

The challenge is therefore to move revenue assurance from testing an exported spreadsheet to interrogating the business itself: continuous, explainable, robust to manipulation, and operable by an AI agent under strict independence.

CORE QUESTION

How much of a real statutory revenue audit can an AI agent run end to end, against a business it cannot alter and on evidence it cannot trust, while still producing the workpaper an auditor would sign?